NOT MEASUREMENT SENSITIVE
MIL-P-85856/21
7 December 1988

MILITARY SPECIFICATION SHEET

POWDER MATERIALS FOR FLAME AND PLASMA SPRAYED COATINGS NICKEL-CHROMIUM-ALUMINUM, TYPE I, CLASS 2, COMPOSITION V

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the sleeving described herein shall consist of this document and the latest issue of MIL-P-85856.

Chemical Composition (1)

Cr	Ni	Fe	Al	C	Mn	Si
15.5 21.5	69.5 min.	1.5 max.	4.5 7.5	0.3 max.	3.0 max.	2.0 max.

(1) Percentages by weight are after removal of binder. Powder shall be a composite of not less than 96.5 percent by weight of an aggregate of nickel-chromium and aluminum powders with not more than 3.5 percent by weight of phenolic binder material capable of volatilizing completely at 1000°F (538°C).

Particle Size/Distribution

+100 Mesh (+150 µm)	1.0 max.
+120 Mesh (+125 µm)	10.0 max.
-325 Mesh (-45 µm)	20.0 max.

Metallographic Inspection

Powder particles shall be capable of exhibiting a composite structure of nickel-chromium alloy core essentially surrounded by aluminum.

AMSC N/A FSC 3439
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Sprayed Coating Properties

Bond Strength	31.0 MPa (4500 psi) min.
Hardness	Rockwell 15T, 90 min.
Microstructure	See paragraph 4.5.4.1.2 Porosity - 15% max. Unmelted Particles - 15% max.

Deposit Efficiency and Deposition Rate

65% min for N_2/H_2 @ 8.0 lb/hr
55% min for A/H_2 @ 8.0 lb/hr

Comparable Documents

Powder which meets the requirements specified is qualified under the following specifications.

General Electric Company	B50TF119, Class A
Pratt & Whitney Aircraft	PWA 1347
Rolls-Royce Ltd.	MSRR 9507/14

Custodians:
Air Force - 20
Navy - AS
Army - MR

Review activities:
Air Force - 99, 71
Navy - SH

Preparing activity:
Navy - AS
(Project No. 3439-0663/21)

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